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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	LetUsCook
Selected Domain	Domain 2: Orbyte Smart SME Challenge
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	SupplyWise AI
Name (Leader)	Chew Jee Syuen
Email & Phone (Leader)	chewjeesyuen@gmail.com +60 11-2075 2580
University Affiliation(s)	Universiti Malaya (UM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Food and beverage (F&B) micro, small, and medium enterprises (MSMEs) in Malaysia operate under severe margin pressures. According to the Department of Statistics Malaysia (DOSM) and SME Corp Malaysia, food service micro-enterprises average single-digit net profit margins (typically 5% to 8%), leaving them highly vulnerable to input cost fluctuations.

In daily operations, kitchen procurement is plagued by fragmented, unstructured communication. Managers coordinate replenishment across dozens of disjointed WhatsApp chats, email threads, and handwritten supplier lists. This lack of standardization generates three systemic operational bottlenecks:

1. **Unstructured Communication Chaos**: Wholesalers quote prices informally via WhatsApp messages and photos. Manually reconciling these quotes takes 4 to 6 hours weekly, leading to frequent transcription errors and missed price changes.
2. **Hidden Landed Costs**: Standard wholesale price lists omit distance-based freight surcharges, tiered delivery thresholds, and strict minimum order quantities (MOQ), causing operators to miscalculate true ingredient unit costs.
3. **Perishable Food Waste**: According to SWCorp Malaysia, commercial food waste accounts for thousands of tonnes daily. Studies by the Khazanah Research Institute and UNEP indicate that sub-optimal procurement synchronization and over-ordering in F&B operations result in a 15% to 20% perishable ingredient spoilage rate.

Compounded by spot-buying markups during unexpected shortages, small operators lose critical margin. Lacking enterprise ERP software, F&B MSMEs require an intelligent, automated solution that eliminates busywork while keeping managers firmly in control.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SupplyWise AI is an intelligent inventory optimization and autonomous supplier discovery platform engineered specifically for independent F&B operators and culinary micro-enterprises within the Orbyte ecosystem.

SupplyWise AI eliminates procurement chaos through four core technical capabilities:

1. **AI Unstructured Quotation Ingestion**: Powered by an LLM-based parsing engine hosted on RunPod, the system ingests unstructured WhatsApp messages, email threads, and PDF price sheets from suppliers. It automatically extracts and standardizes raw text into structured catalog records (SKU pricing, volume discounts, MOQs, and delivery lead times), eliminating hours of manual cross-referencing.
2. **Predictive Inventory Health**: Synchronizing with Orbyte POS sales feeds, the platform tracks ingredient burn rates and projects stockout horizons against dynamic safety thresholds, transforming reactive ordering into proactive replenishment before kitchen operations are disrupted.
3. **Multi-Factor Landed Cost & Proximity Modeling**: Rather than evaluating base wholesale rates in isolation, our algorithmic engine models true unit economics by factoring base prices, road transport proximity (km from Klang Valley distribution hubs), and supplier MOQ fulfillment.
4. **Transparent Decision-Support & Human-in-the-Loop Procurement**: When replenishment is triggered, the system scores and ranks compatible suppliers across three transparent operational priorities:
    - **Lowest Landed Cost**: Maximizing volume margins for planned batch replenishment.
    - **Nearest Proximity**: Minimizing transport food miles, delivery carbon footprint, and transit degradation.
    - **Highest Reliability**: Prioritizing verified vendors with consistent on-time fulfillment and quality ratings.

Managers review clear quotation cards with itemized landed-cost breakdowns and explainable compatibility scores. With one click, the manager approves a finalized purchase order, which is automatically dispatched to the vendor via WhatsApp or email, updating the immutable audit log.

3. Technical Architecture & Implementation Plan

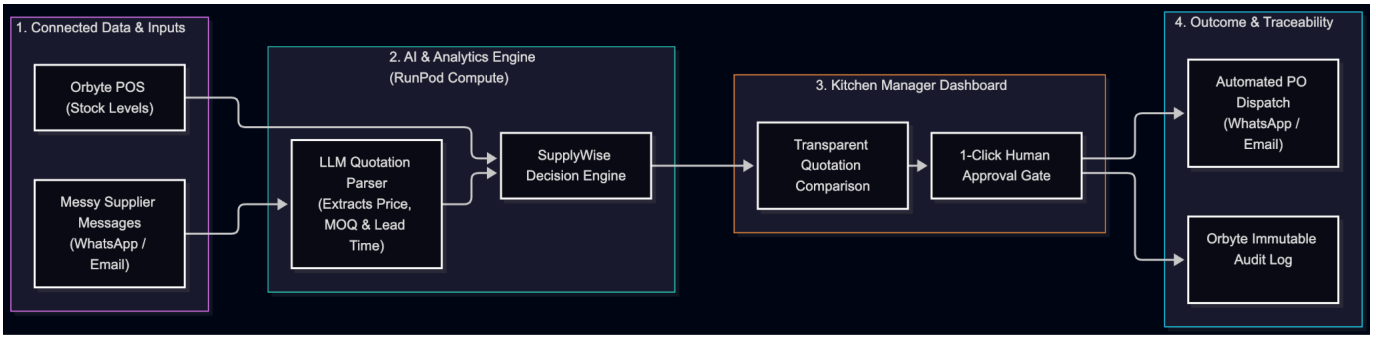
3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technologies Selected
Frontend Framework	React 19, TypeScript 5, Vite 6
Styling & UI Components	Tailwind CSS v4, Lucide Icons
State Management	Zustand
AI Compute Engine	RunPod Serverless / Cloud GPU
Backend & Algorithms	Python 3.11, FastAPI
Database & Persistence	Supabase (PostgreSQL)
Staging	Replit

Block Diagram attached here:



Our team used **RunPod** (SDC sponsor-provided platform) as our primary compute engine hosting LLM models for parsing unstructured WhatsApp/email quotes and catalog vector embeddings.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied human-centered design-thinking to ensure SupplyWise AI is technically viable, intuitive, and ready for Phase 2 deployment:

1. **Empathize & Define**: Research from the World Bank (*Malaysia's Digital Economy*) and MDEC indicates over 60% of Malaysian MSMEs reject conventional ERPs due to interface complexity, cost, and steep training curves, preferring familiar mobile tools like WhatsApp. We defined our goal: build zero-training procurement intelligence that integrates into existing routines.
2. **Ideate:** We designed streamlined quotation comparison cards that standardize distinct supplier quotes into normalized metrics (base price, transport fees, MOQ fulfillment) rather than overwhelming operators with multi-level enterprise menus.
3. **Prototype & Test:** We built mobile-responsive supplier comparison cards with transparent landed-cost breakdowns and explainable compatibility scores, targeting a sub-30-second review and approval workflow for kitchen managers. The core interaction flow was verified through rapid interactive walkthroughs.
4. **Deployment Readiness**: Our architecture connects to Orbyte POS via lightweight REST hooks while offloading heavy unstructured quotation extraction to RunPod serverless endpoints. This modular separation guarantees high client responsiveness on inexpensive kitchen tablets, data schema compatibility, and immediate pilot viability for Phase 2 without requiring merchants to change suppliers or install complex infrastructure.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	*Your Response*
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 8.2**: Achieve higher levels of economic productivity through diversification, technological upgrading, and innovation. **Target 8.3:** Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity, and the growth of micro-, small-, and medium-sized enterprises.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysian F&B SME managers currently spend an average of 4 to 6 hours per week manually cross-checking physical inventory against unstructured WhatsApp messages, supplier phone calls, and paper price lists.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% reduction in weekly procurement administrative hours, tracked via order creation session timestamps and audit logs.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	>80% reduction in routine reordering time, cutting procurement overhead down to under 45 minutes per week.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: N/A

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19	Code (Framework)	MIT
TypeScript 5	Code (Language/Tooling)	Apache 2.0
Vite 6	Code (Bundler/Tooling)	MIT
Tailwind CSS v4	Code (Styling Framework)	MIT
Zustand 5	Code (State Management Library)	MIT
Lucide React	Code (Iconography Library)	ISC
Framer Motion	Code (Animation Library)	MIT
Recharts	Code (Data Visualization Library)	MIT

FastAPI	Code (Backend Web Framework)	MIT
Qwen/Qwen2.5-7B-Instruct	Model (Pre-trained LLM)	Apache License 2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns: N/A

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25 Oct 2026